

Progress in Child Nutrition in India, 2015-2024: Evidence from National Family Health Survey

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Abstract

This paper analyses the progress in child nutrition in India during the period 2015-2024 based on the data available from the latest, 2023-2024, round of the National Family Health Survey. The analysis is based on the concept of child nutrition surface and associated composite index of child nutrition proposed by Chaurasia (2025). The analysis reveals that progress towards child nutrition in the country remains slow and the progress is inconsistent across states/Union Territories and there is high degree of imbalance in the progress in different dimensions of child nutrition in the country and in different states and Union Territories of the country. Nearly all the progress in child nutrition in recent years has been confined to the linear growth dimension of child nutrition. The analysis emphasises the need of a decentralised approach to planning, implementing and monitoring child nutrition improvement efforts to address the local context of child nutrition which is quite pervasive to hasten the progress in child nutrition.

Introduction

The most recent, 2023-2024 round of the National Family Health Survey (NFHS-6) in India suggests that child nutrition continues to be the Achilles heel in the development diaspora of the country. According to the first hand estimates available from the survey, almost 30 per cent children below 5 years of age in the country were low height-for-age (Stunted), close to one third were low weight-for-age (underweight) and around 19 per cent were low weight-for-height (wasted) circa 2024 (Government of India, 2026) despite almost five decades of efforts from the launch of the Integrated Child Development Services (ICDS) in 1975 through the expansion of the Mid-Day Meal/PM Poshan Scheme to the launch of the ambitious Poshan Abhiyaan in 2018. Although the prevalence of stunting, underweight and wasting has decreased over time yet, in terms of absolute numbers, the challenge of under nutrition in the country remains mindboggling. According to the projections prepared by the United Nations, there were around 114 million children below 5 years of age in India in 2024 (United Nations, 2024). This means that an estimated 33.3 million children under 5 years of age in the country were stunted, 36.1 million were underweight, and 21.6 million were wasted. During the 8 years between 2015-2016 and 2023-2024, the number of stunted children in the country decreased by around 14.5 million, number of underweight children decreased by around 8.7 million, but number of wasted children decreased by only around 4.5 million.

There is, however, no single metric to measure child nutrition that can be used to monitor the progress and rank populations in terms of the nutritional status of children. The three indicators of child under nutrition widely used to assess the nutritional status of children – prevalence of stunting, prevalence of underweight and prevalence of wasting – depict the three different dimensions of child nutrition. Stunting reflects linear growth faltering and is a marker of chronic under nutrition (Isanaka et al, 2019; Benjamin-Chung, 2023). Wasting reflects ponderal growth faltering and is a marker of acute under nutrition (Mertens et al, 2023). Underweight reflects faltering in body mass growth which reflects both linear growth faltering and ponderal growth faltering but fails to distinguish between the two. Underweight also reflects thinness, which is also a marker of poor nutritional status, although a thin child may not necessarily be under nourished (Chaurasia, 2017). However, neither stunting nor underweight nor wasting fully captures the multidimensional nature of child nutrition (Svedberg, 2000; Gausman et al, 2022). In India, progress in the three dimensions of child nutrition has been different as the evidence available from different rounds of the National Family Health Survey show. The decrease in the prevalence of stunting has been relatively the fastest but the decrease in the prevalence of wasting has been only marginal. This means that choice of the indicator decides the progress in child nutrition.

In an attempt to construct a single metric of child under nutrition, Svedberg (2000) has proposed the composite index of anthropometric failure (*CIAF*) which has been further modified by Nandy et al (2005). Chaurasia (2017) has shown that classification approaches adopted by Svedberg (2000) and by Nandy et al. (2005) are special cases of the more general classification approach. *CIAF* has widely been used to measure and monitor nutritional status of children, but it lacks specificity as it does not reveal how linear growth faltering, ponderal growth faltering and body mass growth faltering influence *CIAF* (Bose and Mandal, 2010; Al-Sadeeq et al, 2018). *CIAF* assigns equal weight to the three dimensions of child growth – linear growth, ponderal growth and body mass growth – in its construction which is not appropriated as the prevalence of faltering in the three dimensions are different. Ideally, a higher weight should be assigned to that dimension of child growth which has a higher prevalence of faltering. Another problem of *CIAF* is that it is also not possible to estimate the contribution of the change in any of the three dimensions of child growth to the change in *CIAF*. Calculation of *CIAF* requires multidimensional classification of every child in terms of its weight and height relative to the weight and height of the reference child. It cannot be calculated from the prevalence rates of stunting, underweight and wasting that are commonly available through household surveys such as the National Family Health Survey in India or the Demographic and Health Surveys that have been carried out in more than 85 countries. Since *CIAF* cannot tell the relative dominance of stunting, wasting and underweight in deciding the burden of child under nutrition in the population, it is hardly used in tracking the progress in child nutrition and in ranking populations.

Recently Chaurasia (2025) has proposed a composite index of child nutrition (*CICN*) that addresses many of the shortcomings of *CIAF*. The *CICN* is based on the concept of child nutrition surface which recognises that child nutrition is at least a three-dimensional construct which can be described by a three-dimensional vector with linear growth, ponderal growth and body mass growth its three dimensions. The *CICN* proposed by Chaurasia (2025) is based on the area of the child nutrition surface. An advantage of *CICN* is that it is decomposable so that it is possible to ascertain the dominance of the three

dimensions of child nutrition in deciding the nutritional status of the population. It is also possible to decompose the change over time or difference across populations in *CICN* to the change or the difference attributed to the three dimensions of child nutrition. Using the data from the second, third, fourth and fifth rounds of the National Family Health Survey, Chaurasia (2025) estimated that the progress towards child nutrition in India has been slow as the *CICN* in the country increased from 0.609 during 1998-1999 to 0.708 during 2019-2021 and the progress has varied markedly across states and Union Territories of the country. There are four states/Union Territories where *CICN* has decreased during this period. It has also been observed that most of the progress in child nutrition in the country has been confined to the linear growth dimension while there has been little progress in the ponderal growth dimension of child nutrition.

This paper has four objectives. The first is to estimate *CICN* for India and its states and Union Territories for the period 2023-2024 based on the estimates of the prevalence of stunting, underweight and wasting available from the latest, round of the National Family Health Survey, 2023-2024 (NFHS-6). The second objective is to analyse the change in the rank of states/Union Territories in *CICN* during 2023-2024 compared to the rank during 2019-2021 and 2015-2016. The third objective is to decompose the change in *CICN* between 2015-2016 and 2023-2024 in terms of the change in the proportion of children who are not stunted, proportion who are not underweight and proportion who are not wasted. Finally, the fourth objective of the paper is to analyse how rural-urban difference in the proportion of children not stunted, not underweight and not wasted contributes to rural-urban difference in *CICN* in India and states. The analysis has been carried out for the country and for its states and Union Territories except Manipur as data for the state are not available from NFHS-6. District level estimates of the prevalence of stunting, underweight and wasting are also currently not available from NFHS-6 so that district level analysis is not possible at the moment.

Prevalence of Stunting, Underweight and Wasting

Estimates of the proportion of children under 5 years of age who are low height-for-age (stunted), low weight-for-age (underweight) and low weight-for-height (wasted) in India and in its 36 states and Union Territories are given in table 1 for the period 2015-2016, 2019-2021 and 2023-2024. Estimates the prevalence of stunting, underweight and wasting for Manipur are not available for the period 2023-2024 as the 2023-2024 round of the National Family Health Survey (NFHS-6) could not be conducted in the state. The proportion of children under 5 years of age who are low height-for-age decreased in the country from 38.4 per cent during 2015-2016 to 29.3 per cent during 2023-2024 while the proportion of children under 5 years of age who are low weight-for-age decreased from 35.8 per cent to 31.8 per cent during this period. The proportion of children under 5 years of age who are low weight-for-height, however, decreased from 21 per cent to 19 per cent between 2015-2016 and 2023-2024 because of only a marginal decrease from 19.3 per cent to 19 per cent between 2019-2021 and 2023-2024. The progress in child nutrition in the country has been different in terms of linear growth, body mass growth and ponderal growth. It has been relatively the fastest in the linear growth dimension while there has been near stagnation in the progress in the ponderal growth dimension of child nutrition.

Table 1: Proportion of children under 5 years of age who were low height-for-age (stunted), low weight-for-age (underweight) and low weight-for-height (wasted) in India and states/Union Territories, 2015-2024.

Country/State/Union Territory	Proportion (per cent) of children under 5 years of age who are								
	Low height-for-age (Stunted)			Low weight-for-age (Underweight)			Low weight-for-height (Wasted)		
	2023-2024	2019-2021	2015-2016	2023-2024	2019-2021	2015-2016	2023-2024	2019-2021	2015-2016
India	29.3	35.5	38.4	31.8	32.1	35.8	19.0	19.3	21.0
Andaman & Nicobar Islands	26.7	22.5	23.3	19.3	23.6	21.6	11.1	16.0	18.9
Andhra Pradesh	24.6	31.2	31.4	23.2	29.6	31.9	12.3	16.1	17.2
Arunachal Pradesh	24.1	28.0	29.4	18.2	15.4	19.4	16.1	13.1	17.3
Assam	30.3	35.3	36.4	30.7	32.8	29.8	21.4	21.7	17.0
Bihar	35.6	42.9	48.3	35.7	41.0	43.9	19.0	22.9	20.8
Chandigarh	19.0	25.3	28.7	31.6	20.6	24.5	19.5	8.4	10.9
Chhattisgarh	27.5	34.6	37.6	34.7	31.3	37.7	21.1	18.9	23.1
Dadra & Nagar Haveli Daman & Diu	37.1	39.4	31.9	41.3	38.7	27.0	24.2	21.6	15.9
Delhi	21.4	26.9	37.2	14.5	21.0	35.8	10.6	19.0	26.7
Goa	19.4	25.8	20.1	19.9	24.0	23.8	16.3	19.1	21.9
Gujarat	35.3	39.0	38.5	35.5	39.7	39.3	20.2	25.1	26.4
Haryana	25.9	27.5	34.0	28.9	21.5	29.4	16.6	11.5	21.2
Himachal Pradesh	20.6	30.8	26.3	16.8	25.5	21.2	10.4	17.4	13.7
Jammu & Kashmir	21.4	26.9	27.4	14.5	21.0	16.6	10.6	19.0	12.2
Jharkhand	35.0	39.6	36.2	41.1	39.4	35.2	22.3	22.4	26.1
Karnataka	26.5	35.4	19.7	27.8	32.9	16.1	18.7	19.5	15.7
Kerala	20.1	23.4	30.9	17.8	19.7	18.7	10.9	15.8	9.3
Ladakh	26.0	30.5	26.8	14.5	20.4	23.6	10.6	17.5	13.7
Lakshadweep	34.1	32.0	42.0	25.9	25.8	42.8	12.4	17.4	25.8
Madhya Pradesh	31.4	35.7	34.4	39.7	33.0	36.0	23.8	18.9	25.6
Maharashtra	29.5	35.2	28.9	31.4	36.1	13.8	19.9	25.6	6.8
Manipur		23.4	28.9		13.3	13.8		9.9	6.8
Meghalaya	36.8	46.5	43.8	25.3	26.6	28.9	13.2	12.1	15.3
Mizoram	26.1	28.9	28.1	14.5	12.7	12.0	8.1	9.8	6.1
Nagaland	25.1	32.7	28.6	12.2	26.9	16.7	8.9	19.1	11.3
Odisha	26.8	31.0	34.1	31.6	29.7	34.4	22.1	18.1	20.4
Puducherry	26.4	30.9	23.7	27.4	21.8	22.0	15.0	11.2	23.6
Punjab	20.4	24.5	25.7	23.7	16.9	21.6	18.0	10.6	15.6
Rajasthan	29.6	31.8	39.1	33.3	27.6	36.7	19.8	16.8	23.0
Sikkim	20.1	22.3	29.6	11.2	13.1	14.2	9.4	13.6	14.2
Tamil Nadu	20.7	25.0	27.1	23.2	22.0	23.8	17.4	14.6	19.7
Telangana	27.0	33.1	28.0	27.8	31.8	28.4	16.9	21.7	18.1
Tripura	21.9	33.3	24.3	25.7	25.6	24.1	18.4	18.2	16.8
Uttar Pradesh	31.5	39.7	46.3	34.5	32.1	39.5	19.2	17.3	17.9
Uttarakhand	20.0	27.0	33.5	19.6	21.0	26.6	11.0	13.2	19.5
West Bengal	22.4	33.8	32.5	28.5	32.2	31.6	20.3	20.3	20.3

Remarks: NFHS-6 was not conducted in Manipur.

Source: Government of India (2017; 2022; 2026a).

Among different states and Union Territories of the country, the progress in all the three dimensions of child nutrition varies widely. The prevalence of stunting was relatively the highest in Bihar but relatively the lowest in Karnataka during 2015-2016 but in Meghalaya and Sikkim respectively during 2019-2021 and respectively in Dadra & Nagar Haveli and Daman & Diu and Chandigarh during 2023-2024. The prevalence of underweight, on the other hand, was relatively the highest in Bihar but relatively the lowest in Mizoram during 2015-2016 and 2019-2021 but respectively in Dadra & Nagar Haveli and Daman & Diu and Sikkim during 2023-2024. Finally, the prevalence of wasting was relatively the highest in Delhi but relatively the lowest in Mizoram during 2015-2016 but respectively in Maharashtra and Chandigarh during 2019-2021 and Dadra & Nagar Haveli and Daman & Diu and Mizoram during 2023-2024. Dadra & Nagar Haveli and Daman & Diu has the dubious distinction of having relatively the highest prevalence of stunting, underweight and wasting in children under 5 years of age across the states and Union Territories during the period 2023-2024. The across states/Union Territories variation in all the three prevalence rates decreased during 2019-2021 relative to 2015-2016 but increased during 2023-2024 relative to 2019-2021.

Composite Index of Child Nutrition

Details of the construction of *CICN* are given elsewhere (Chaurasia, 2025). *CICN* is based on the concept of child nutrition surface and is defined as

$$CICN = \frac{\sqrt{ns \times nu} + \sqrt{nu \times nw} + \sqrt{nw \times ns}}{3} \quad (1)$$

where *ns* is the proportion of children under 5 years of age who are not low height-for-age, *nu* is the proportion not low weight-for-age and *nw* is the proportion not low weight-for-height. *CICN* ranges from 0 to 1 and the higher the *CICN* the better the nutritional status of children and vice versa. When there is no child who is either stunted or underweight or wasted, *CICN* is equal to 1. *CICN* serves as a convenient metric for monitoring the progress and ranking populations in child nutrition. It is a composite population-level metric constructed from the prevalence of children who are low height-for-age, low weight-for-age and low weight-for-height. It is not a direct measure of the nutritional status of a child. It also does not capture the joint occurrence different forms of undernutrition in the same child. The appropriate index for assessing the nutritional status of a child is the *CIAP* proposed by Svedberg (2000), its modification (Nandy et al, 2005) or its generalization (Chaurasia, 2017). However, application of *CIAP* requires child-specific data. When only prevalence rates of three forms of under-nutrition are available *CICN* a broad, macro, perspective of the challenge of undernutrition among children in the population that can be used for ranking populations, for comparing populations and for monitoring progress towards child nutrition. Being a composite metric, *CICN* has some limitations. Ranking of populations, comparison across populations and monitoring progress based on *CICN* may be different from ranking, comparison and monitoring in terms of the three dimensions of child nutrition. It is, therefore, necessary that the change over time or the difference across populations in *CICN* is decomposed in terms of the change or the difference in the three dimensions of child nutrition to fully understand the complex dynamics of the nutrition status of children in the population.

CICN is a mathematical transformation of *ns*, *nu* and *nw*. The uncertainty or the error in the estimation of *CICN*, therefore, is determined by the uncertainty or the error in the estimation of *ns*, *nu* and *nw* or in *s*, *u* and *n* and may be estimated through the application of the law of the propagation of uncertainty (Taylor and Kuyatt, 1994). Assuming that *ns*, *nu* and *nw* are mutually independent, the uncertainty or the error associated in the estimation of *CICN* may be estimated as

$$\sigma_{CICN} = \sqrt{\frac{(\sqrt{nu} + \sqrt{nw})^2}{36 \times ns} \times \sigma_{ns}^2 + \frac{(\sqrt{ns} + \sqrt{nw})^2}{36 \times nu} \times \sigma_{nu}^2 + \frac{(\sqrt{ns} + \sqrt{nu})^2}{36 \times nw} \times \sigma_{nw}^2} \quad (2)$$

Assuming that σ_{CICN} is a reliable estimate of the standard deviation of *CICN*, the upper and lower confidence limits of *CICN* may be obtained. Calculation of the confidence interval of *CICN*, therefore, requires measurement of the uncertainty in the estimation of *ns*, *nu* and *nw*. If estimates of uncertainty associated with *ns*, *nu* and *nw* are not available, confidence limits of *CICN* may not be calculated, although confidence limits are important testing the rank robustness of *CICN*. If confidence limits overlap, then variation in *CICN* is usually noise. It is therefore customary to construct categories rather than relying entirely on exact value of *CICN*.

The change or the difference in *CICN* may be decomposed into the change or the difference in *ns*, *nu* and *nw* following the logarithmic mean divisia index (LMDI) decomposition approach (Ang and Liu, 1998; Ang, 2015). This approach is based on the logarithmic mean (Carlson, 1972) instead of the arithmetic mean. Unlike the arithmetic mean, logarithmic means is more balanced proportional middle ground and is more suitable when the indicator being decomposed is a proportion variable as is the case here. The logarithmic mean of two numbers is greater than their geometric mean but less than their arithmetic mean. The change or the difference in *CICN* may be written as

$$\nabla_{CICN} = CICN_2 - CICN_1 = \frac{\sqrt{ns_2 * nu_2} + \sqrt{nu_2 * nw_2} + \sqrt{nw_2 * ns_2}}{3} - \frac{\sqrt{ns_1 * nu_1} + \sqrt{nu_1 * nw_1} + \sqrt{nw_1 * ns_1}}{3}$$

or

$$\nabla_{CICN} = \frac{\sqrt{ns_2 * nu_2} - \sqrt{ns_1 * nu_1}}{3} + \frac{\sqrt{nu_2 * nw_2} - \sqrt{nu_1 * nw_1}}{3} + \frac{\sqrt{nw_2 * ns_2} - \sqrt{nw_1 * ns_1}}{3} \quad (3)$$

Using the logarithmic mean, we can write

$$\sqrt{ns_2 * nu_2} - \sqrt{ns_1 * nu_1} = \frac{\sqrt{ns_2 * nu_2} - \sqrt{ns_1 * nu_1}}{\ln\left(\frac{\sqrt{ns_2 * nu_2}}{\sqrt{ns_1 * nu_1}}\right)} * \ln\left(\frac{\sqrt{ns_2 * nu_2}}{\sqrt{ns_1 * nu_1}}\right) = L_{lb} * \ln\left(\frac{\sqrt{ns_2 * nu_2}}{\sqrt{ns_1 * nu_1}}\right)$$

or

$$\sqrt{ns_2 * nu_2} - \sqrt{ns_1 * nu_1} = \frac{L_{lb}}{2} * \ln\left(\frac{ns_2}{ns_1}\right) + \frac{L_{lb}}{2} * \ln\left(\frac{nu_2}{nu_1}\right) \quad (4)$$

where

$$L_{lb} = \frac{\sqrt{ns_2 * nu_2} - \sqrt{ns_1 * nu_1}}{\ln\left(\frac{\sqrt{ns_2 * nu_2}}{\sqrt{ns_1 * nu_1}}\right)} \quad (5)$$

is the logarithmic mean of $\sqrt{ns_2 * nu_2}$ and $\sqrt{ns_1 * nu_1}$ (Carlson, 1972). Similarly, we can also write

$$\sqrt{nu_2 * nw_2} - \sqrt{nu_1 * nw_1} = \frac{L_{bp}}{2} * \ln\left(\frac{nu_2}{nu_1}\right) + \frac{L_{bp}}{2} * \ln\left(\frac{nw_2}{nw_1}\right) \quad (6)$$

and

$$\sqrt{nw_2 * ns_2} - \sqrt{nw_1 * ns_1} = \frac{L_{pl}}{2} * \ln\left(\frac{nw_2}{nw_1}\right) + \frac{L_{pl}}{2} * \ln\left(\frac{ns_2}{ns_1}\right) \quad (7)$$

so that

$$\nabla_{CICN} = \frac{(L_{lb}+L_{pl})}{6} * \ln\left(\frac{ns_2}{ns_1}\right) + \frac{(L_{lb}+L_{bp})}{6} * \ln\left(\frac{nu_2}{nu_1}\right) + \frac{(L_{pl}+L_{bp})}{6} * \ln\left(\frac{nw_2}{nw_1}\right) \quad (8)$$

$$\nabla_{CICN} = C_{ns} + C_{nu} + C_{nw} \quad (9)$$

where C_{ns} is the contribution of the change in the proportion of children not stunted to the change in $CICN$. Similarly, C_{nu} is the contribution of the change in the proportion of children who are not underweight and C_{nw} is the contribution of the change in the proportion of children who are not wasted to the change in $CICN$.

Notice that if prevalence rates are available for more than two points in time, say at three points in time, then

$$\ln\left(\frac{ns_3}{ns_1}\right) = \ln\left(\frac{ns_3}{ns_2}\right) + \ln\left(\frac{ns_2}{ns_1}\right)$$

so that the change in $CICN$ between time 1 and time 3 can be decomposed into the change between time 1 and time 2 and the change between time 2 and time 3 as

$$\begin{aligned} \nabla_{CICN} &= \frac{(L_{lb}+L_{pl})}{6} * \ln\left(\frac{ns_3}{ns_2}\right) + \frac{(L_{lb}+L_{pl})}{6} * \ln\left(\frac{ns_2}{ns_1}\right) + \\ &\frac{(L_{lb}+L_{bp})}{6} * \ln\left(\frac{nu_3}{nu_2}\right) + \frac{(L_{lb}+L_{bp})}{6} * \ln\left(\frac{nu_2}{nu_1}\right) + \\ &\frac{(L_{pl}+L_{bp})}{6} * \ln\left(\frac{nw_3}{nw_2}\right) + \frac{(L_{pl}+L_{bp})}{6} * \ln\left(\frac{nw_2}{nw_1}\right) \end{aligned} \quad (10)$$

One problem with the LMDI decomposition approach is the handling zero values in the data as the logarithm of zero is undefined, and division by zero creates mathematical errors (Wood and Lenzen, 2006). Although, in surveys based on large samples, it is rare that the prevalence of under nutrition in any dimension of child nutrition is zero, yet this may be a problem in small scale surveys or in small geographical areas and in subgroups of population. Two alternatives have been suggested to address the zero-value problem in the LMDI decomposition approach. The first is the analytical limits approach while the second is the small-value substitution approach (Ang and Liu, 2007). The analytical limits approach is a mathematically robust approach. It uses the mathematical limits of the LMDI function as data points approach zero. When one value is zero, the logarithmic mean weight becomes zero and, even though the log-ratio tends toward infinity, the product of the weight and the log-ratio mathematically converges to exactly zero (Wood and Lenzen, 2006). The small-value substitution approach is the numerical approach. It suggests replacing all zero values in the dataset with a tiny positive number. The first approach is recommended because if the dataset contains a large number of zero values or very small real values, replacing zero by a tiny positive number may lead to significant calculation errors (Wood and Lenzen, 2006).

Child Nutrition in India, 2015-2024

Table 2 gives estimates of *CICN* for India and for its states and Union Territories for the period 2015-2016, 2019-2021 and 2023-2024. These estimates are based on the prevalence of stunting, underweight and wasting obtained from different rounds of the National Family Health Survey given in table 1. Based on the *CICN*, the nutritional status of children in a population may be termed as very good if $CICN \geq 0.850$; good if $0.800 \leq CICN < 0.850$; average if $0.750 \leq CICN < 0.800$; poor if $0.700 \leq CICN < 0.750$; and very poor if $CICN < 0.700$. These cutoff points have been arrived at on the basis of quantiles of the distribution of proportion of children not stunted, proportion of children not underweight, and proportion of children not wasted across states and Union Territories of the country during the period 2023-2024 as obtained from NFHS-6. For India as a whole, *CICN* is estimated to be 0.731 during the period 2023-2024 which means that nutritional status of children in India remains poor.

CICN varies widely across states and Union Territories. Sikkim with a $CICN > 0.850$ is the only state/Union Territory where the nutritional status of children may be termed as very good. In 11 states/Union Territories, nutritional status of children may be termed as good as $0.800 \leq CICN < 0.850$ in these states/Union Territories and average ($0.750 \leq CICN < 0.800$) in 11 states/Union Territories. There are 12 states/Union Territories where the nutritional status of children may be termed as either poor or poor as *CICN* is less than 0.750 in these states/Union Territories. There are 5 states/Union Territories which may be termed as hotspot states/Union Territories of the country in terms of child nutrition as *CICN* is below 0.700 in these states/Union Territories.

Table 2 shows the trend in *CICN* in India and its constituent states and Union Territories during the period 2015-2024. The annual rate of increase in *CICN* has been faster during the period 2019-2024 compared to the period 2015-2021 which implies that the progress in child nutrition in the country accelerated during the period 2019-2024 compared to the period 2015-2021. Among different states and Union Territories, the progress in child nutrition as reflected through the trend in *CICN* has been mixed. In 14 states/Union Territories, annual rate of increase in *CICN* during 2019-2024 was slower than that during 2015-2021 and in 9 states/Union Territories, *CICN* decreased, instead increased, during this period, indicating a deterioration in the nutritional status of children. During 2015-2021 also *CICN* decreased in 14 states/Union Territories. However, in 13 of these states/Union Territories, *CICN* increased during 2019-2024. Dadra & Nagar Haveli and Daman & Diu is the only state/Union Territory in the country in which *CICN* decreased during 2015-2021 as well as during 2019-2024.

The pace of increase in *CICN* has also been different in different states and Union Territories the country during 2019-2024 and during 2015-2021. During 2019-2024, the increase in *CICN* has been the most rapid in Nagaland followed by Delhi and Himachal Pradesh. During 2015-2021, the increase in *CICN* was the most rapid in Lakshadweep followed by Delhi. Combining the two period, the increase in *CICN* during 2015-2024 has been the most rapid in Delhi. On the other hand, the decrease in *CICN* has been the most rapid in Chandigarh during 2019-2024 and in Maharashtra during 2015-2021. During 2015-2024, the decrease in *CICN* has been the most rapid in Maharashtra because of the very rapid decrease during 2015-2021.

Table 2: *CICN* in India and states/Union Territories, 2015-2024.

Country/State/Union Territory	<i>CICN</i>			Average annual change (per cent)		
	2023-2024	2019-2021	2015-2016	2019-2024	2015-2021	2015-2024
India	0.731	0.708	0.680	0.675	0.629	0.649
Andaman & Nicobar Islands	0.808	0.792	0.787	0.383	0.132	0.258
Andhra Pradesh	0.798	0.741	0.729	1.422	0.302	0.862
Arunachal Pradesh	0.805	0.810	0.778	-0.161	0.627	0.331
Assam	0.724	0.699	0.719	0.639	-0.513	0.063
Bihar	0.696	0.639	0.615	1.409	0.611	1.010
Chandigarh	0.765	0.817	0.784	-1.737	0.662	-0.238
Chhattisgarh	0.721	0.715	0.669	0.191	0.911	0.641
Dadra & Nagar Haveli Daman & Diu	0.655	0.664	0.748	-0.224	-2.111	-1.168
Delhi	0.844	0.776	0.666	2.255	2.200	2.220
Goa	0.815	0.770	0.781	1.115	-0.265	0.425
Gujarat	0.694	0.651	0.651	1.063	0.019	0.541
Haryana	0.761	0.796	0.717	-1.188	1.595	0.552
Himachal Pradesh	0.840	0.753	0.795	2.176	-1.048	0.564
Jammu & Kashmir	0.844	0.776	0.811	1.691	-0.857	0.417
Jharkhand	0.669	0.658	0.674	0.339	-0.308	-0.065
Karnataka	0.756	0.705	0.828	1.276	-3.084	-0.904
Kerala	0.837	0.803	0.800	0.836	0.081	0.459
Ladakh	0.828	0.770	0.785	1.431	-0.360	0.535
Lakshadweep	0.755	0.748	0.628	0.180	2.993	1.587
Madhya Pradesh	0.681	0.705	0.679	-0.796	0.527	0.031
Maharashtra	0.729	0.676	0.831	1.340	-3.882	-1.271
Manipur		0.843	0.831		0.303	
Meghalaya	0.744	0.705	0.699	0.978	0.146	0.562
Mizoram	0.835	0.825	0.842	0.245	-0.417	-0.086
Nagaland	0.844	0.736	0.809	2.692	-1.819	0.436
Odisha	0.731	0.736	0.701	-0.159	0.684	0.368
Puducherry	0.769	0.784	0.769	-0.492	0.299	0.002
Punjab	0.793	0.825	0.790	-1.082	0.713	0.040
Rajasthan	0.723	0.744	0.668	-0.712	1.522	0.684
Sikkim	0.863	0.836	0.804	0.688	0.793	0.740
Tamil Nadu	0.795	0.794	0.764	0.054	0.591	0.390
Telangana	0.760	0.710	0.751	1.245	-1.014	0.116
Tripura	0.780	0.741	0.782	0.962	-1.025	-0.031
Uttar Pradesh	0.714	0.698	0.646	0.510	1.044	0.844
Uttarakhand	0.831	0.795	0.733	1.202	1.230	1.220
West Bengal	0.762	0.710	0.717	1.290	-0.166	0.562

Remarks: NFHS-6 was not conducted in Manipur.

Source: Author

Table 3: Rank of states/Union Territories in terms of *CICN* in 2015-2016, 2019-2021 and 2023-2024.

State/Union Territory	Rank in <i>CICN</i>			
	2023-2024	2019-2021	2015-2016	
Andaman & Nicobar Islands	11	10	10	Decreased
Andhra Pradesh	13	19	21	
Arunachal Pradesh	12	5	15	Decreased
Assam	27	29	22	
Bihar	31	35	35	
Chandigarh	18	4	12	Decreased
Chhattisgarh	29	23	29	Decreased
Dadra & Nagar Haveli Daman & Diu	35	32	19	Decreased
Delhi	2	12	31	
Goa	10	15	14	
Gujarat	32	34	32	
Haryana	20	7	24	Decreased
Himachal Pradesh	5	16	8	
Jammu & Kashmir	2	12	4	
Jharkhand	34	33	28	Decreased
Karnataka	22	28	3	
Kerala	6	6	7	
Ladakh	9	14	11	
Lakshadweep	23	17	34	Decreased
Madhya Pradesh	33	27	27	Decreased
Maharashtra	26	31	2	
Meghalaya	24	26	26	
Mizoram	7	2	1	Decreased
Nagaland	4	21	5	
Odisha	25	22	25	Decreased
Puducherry	17	11	16	Decreased
Punjab	15	3	9	Decreased
Rajasthan	28	18	30	Decreased
Sikkim	1	1	6	
Tamil Nadu	14	9	17	Decreased
Telangana	21	25	18	
Tripura	16	20	13	
Uttar Pradesh	30	30	33	
Uttarakhand	8	8	20	
West Bengal	19	24	23	

Remarks: Ranks are based on 35 states/Union Territories as NFHS-6 was not conducted in Manipur.

Source: Author

Because of differential pace of change, rank of a state/Union Territory vis-a-vis other states/Union Territories in terms of *CICN* has been different in 2023-2024 as compared to that in 2019-2021 and 2015-2016 (Table 3). The rank of 15 states/Union Territories in terms of *CICN* decreased during 2023-2024 compared to the rank during 2019-2021 while

rank of 18 states/Union Territories improved. Sikkim, Kerala, Uttar Pradesh and Uttarakhand are the only four states/Union Territories where the rank in *CICN* during 2019-2021 and during 2023-2024 has remained the same. In some states/Union Territories, the change in the rank has been very marked. For example, Chandigarh ranked 4 in terms of *CICN* during 2019-2021 but only 18 during 2023-2024. Similarly, Haryana ranked 7 during 2019-2021 but 20 during 2023-2024 in terms of *CICN*. On the other hand, Nagaland ranked 21 during 2019-2021 in terms of *CICN* but the rank of the state jumped to 4 during 2023-2024. In Delhi and Jammu & Kashmir also, the rank in *CICN* improved substantially between 2019-2021 and 2023-2024. The decrease in rank of a state/Union Territory indicates that the progress has been slower as compared to other states/Union Territories. Similarly, an increase in the rank of a state/Union Territory in *CICN* indicates a faster progress in child nutrition relative to the progress in other states and Union Territories.

Decomposition of the Change in *CICN*

Table 4 shows how decrease in the proportion of stunted children, underweight children and wasted children has contributed to the increase in *CICN* in India during the period 2015-2024. The evidence available from the latest round of the National Family Health Survey suggests that more than 60 per cent of the increase in *CICN* in the country between 2015-2016 and 2023-2024 is attributed to the decrease in the proportion of stunted (low weight-for-age) children while around 27 per cent is attributed to the decrease in the proportion of underweight (low weight-for-age) children. On the other hand, the decrease in the proportion of wasted (low weight-for-height) children accounted for an increase of around 12 per cent in *CICN*. Table 4 suggests that the main driver of the improvement in the nutritional status of children in the country during the 8 years between 2015-2016 and 2023-2024 has been the decrease in the proportion of stunted children while the decrease in the proportion of wasted children contributed only marginally to the increase in *CICN*.

The contribution of the decrease in the proportion of stunted children, underweight children and wasted children to the increase in *CICN* has, however, been different during the period 2015-2021 and the period 2019-2024. During the period 2015-2021, more than 45 per cent of the increase in *CICN* was attributed to the decrease in the proportion of underweight children while the decrease in the proportion of stunted children accounted for an increase of almost 37 per cent. The rest of the increase in *CICN* was the result of the decrease in the proportion of wasted children. During the period 2019-2024, however, more than 90 per cent of the increase in *CICN* was accounted by the decrease in the proportion of stunted children while the decrease in the proportion of underweight children and the decrease in the proportion of wasted children contributed only marginally to the increase in *CICN*. Unlike the period 2015-2021, the improvement in the nutritional status of children in the country during the period 2019-2024 has almost entirely been because of the decrease in the proportion of stunted (low height-for-age) children. The proportion of underweight children and the proportion of wasted children decreased in the country during this period but the progress in these two dimensions of child nutrition contributed only marginally to the improvement in the nutritional status of children as reflected through the increase in *CICN*.

Table 4: Decomposition of the change in *CICN* during 2015-2024 in India and states/Union Territories.

Country/State/Union Territory	2015-2024				2019-2024				2015-2021			
	Change in <i>CICN</i>	Change in <i>CICN</i> due to			Change in <i>CICN</i>	Change in <i>CICN</i> due to			Change in <i>CICN</i>	Change in <i>CICN</i> due to		
		Change in <i>ns</i>	Change in <i>nu</i>	Change in <i>nw</i>		Change in <i>ns</i>	Change in <i>nu</i>	Change in <i>nw</i>		Change in <i>ns</i>	Change in <i>nu</i>	Change in <i>nw</i>
India	0.052	0.032	0.014	0.006	0.023	0.021	0.001	0.001	0.029	0.011	0.013	0.005
Andaman & Nicobar Islands	0.021	-0.012	0.008	0.025	0.015	-0.015	0.015	0.015	0.005	0.003	-0.007	0.009
Andhra Pradesh	0.069	0.024	0.030	0.015	0.056	0.023	0.022	0.012	0.013	0.001	0.008	0.003
Arunachal Pradesh	0.026	0.019	0.004	0.004	-0.005	0.014	-0.009	-0.009	0.031	0.005	0.013	0.013
Assam	0.005	0.022	-0.003	-0.013	0.026	0.018	0.007	0.001	-0.021	0.004	-0.010	-0.014
Bihar	0.081	0.046	0.029	0.005	0.055	0.025	0.018	0.011	0.026	0.021	0.011	-0.006
Chandigarh	-0.019	0.033	-0.025	-0.027	-0.051	0.021	-0.038	-0.034	0.032	0.012	0.013	0.007
Chhattisgarh	0.051	0.034	0.011	0.006	0.006	0.024	-0.012	-0.007	0.046	0.011	0.022	0.013
Dadra & Nagar Haveli Daman & Diu	-0.093	-0.018	-0.050	-0.025	-0.010	0.009	-0.010	-0.008	-0.084	-0.027	-0.040	-0.017
Delhi	0.178	0.055	0.072	0.051	0.063	0.018	0.020	0.025	0.115	0.037	0.052	0.026
Goa	0.034	0.002	0.013	0.018	0.045	0.022	0.014	0.009	-0.011	-0.020	-0.001	0.009
Gujarat	0.043	0.011	0.013	0.019	0.042	0.013	0.015	0.015	0.001	-0.002	-0.001	0.004
Haryana	0.044	0.028	0.002	0.014	-0.034	0.005	-0.024	-0.015	0.078	0.023	0.026	0.029
Himachal Pradesh	0.045	0.020	0.015	0.010	0.089	0.037	0.030	0.023	-0.044	-0.017	-0.015	-0.012
Jammu & Kashmir	0.033	0.021	0.007	0.005	0.069	0.020	0.022	0.028	-0.036	0.002	-0.015	-0.023
Jharkhand	-0.005	0.004	-0.021	0.012	0.010	0.016	-0.006	0.000	-0.016	-0.012	-0.015	0.011
Karnataka	-0.072	-0.023	-0.039	-0.010	0.056	0.034	0.019	0.003	-0.128	-0.057	-0.059	-0.012
Kerala	0.037	0.039	0.003	-0.005	0.033	0.011	0.006	0.016	0.003	0.027	-0.003	-0.021
Ladakh	0.043	0.003	0.030	0.010	0.058	0.016	0.019	0.022	-0.015	-0.014	0.011	-0.012
Lakshadweep	0.127	0.029	0.059	0.040	0.007	-0.007	0.000	0.014	0.120	0.036	0.059	0.026
Madhya Pradesh	0.002	0.010	-0.013	0.006	-0.023	0.015	-0.023	-0.014	0.026	-0.005	0.010	0.020
Maharashtra	-0.102	-0.002	-0.059	-0.040	0.059	0.021	0.018	0.020	-0.161	-0.024	-0.078	-0.060
Meghalaya	0.045	0.027	0.012	0.006	0.039	0.038	0.004	-0.003	0.006	-0.011	0.008	0.009
Mizoram	-0.007	0.007	-0.008	-0.006	0.010	0.010	-0.006	0.005	-0.017	-0.003	-0.002	-0.012
Nagaland	0.035	0.013	0.015	0.008	0.113	0.029	0.051	0.033	-0.078	-0.016	-0.036	-0.026
Odisha	0.029	0.025	0.010	-0.005	-0.005	0.014	-0.006	-0.012	0.034	0.011	0.016	0.007

PROGRESS IN CHILD NUTRITION IN INDIA

Country/State/Union Territory	2015-2024				2019-2024				2015-2021			
	Change in <i>CICN</i>	Change in <i>CICN</i> due to			Change in <i>CICN</i>	Change in <i>CICN</i> due to			Change in <i>CICN</i>	Change in <i>CICN</i> due to		
		Change in <i>ns</i>	Change in <i>nu</i>	Change in <i>nw</i>		Change in <i>ns</i>	Change in <i>nu</i>	Change in <i>nw</i>		Change in <i>ns</i>	Change in <i>nu</i>	Change in <i>nw</i>
Puducherry	0.000	-0.009	-0.018	0.028	-0.014	0.016	-0.019	-0.011	0.014	-0.025	0.001	0.039
Punjab	0.003	0.018	-0.007	-0.008	-0.032	0.014	-0.022	-0.023	0.035	0.004	0.015	0.015
Rajasthan	0.055	0.033	0.012	0.010	-0.020	0.007	-0.019	-0.009	0.075	0.026	0.031	0.018
Sikkim	0.059	0.034	0.010	0.015	0.027	0.008	0.006	0.013	0.032	0.027	0.004	0.002
Tamil Nadu	0.031	0.022	0.002	0.007	0.002	0.014	-0.004	-0.009	0.030	0.007	0.006	0.016
Telangana	0.009	0.003	0.002	0.004	0.051	0.022	0.014	0.015	-0.042	-0.018	-0.012	-0.012
Tripura	-0.003	0.008	-0.005	-0.005	0.040	0.041	0.000	-0.001	-0.042	-0.033	-0.005	-0.004
Uttar Pradesh	0.068	0.054	0.018	-0.004	0.015	0.028	-0.008	-0.005	0.053	0.026	0.026	0.002
Uttarakhand	0.098	0.047	0.024	0.027	0.035	0.023	0.005	0.007	0.063	0.024	0.019	0.020
West Bengal	0.045	0.034	0.011	0.000	0.052	0.039	0.013	0.000	-0.007	-0.005	-0.002	0.000

Remarks: NFHS-6 was not conducted in Manipur.

Source: Author

The relative contribution of the improvement in the three dimensions of child nutrition to the improvement in the nutritional status of children during the period under reference has varied across the states and Union Territories of the country. There are only 23 states/Union Territories in which the nutritional status of children improved in all three dimensions of child nutrition. In 12 states/Union Territories, progress reversed in at least one dimension of child nutrition so that in 7 states/Union Territories, progress in the nutritional status of children reversed as reflected through the decrease in *CICN*. The progress in child nutrition has been the most rapid in Delhi increase in *CICN* in Delhi where *CICN* increased from 0.666 during 2015-2016 to 0.844 during 2023-2024. More than 40 per cent of the increase in *CICN* in the state was due to the decrease in the proportion of stunted children, and rest of the increase is attributed to the decrease in the proportion of underweight children and the proportion of wasted children. On the other hand, *CICN* decreased, instead increased, the most rapidly in Maharashtra because of the reversal in progress in all the three dimensions of child nutrition. The progress in all the three dimensions of child nutrition also reversed in Dadra & Nagar Haveli and Daman & Diu and Karnataka. In all the three states/Union Territories, more than half of the decrease in *CICN* was due to the increase in the proportion of underweight children. However, the contribution of the increase in the proportion of stunted children and the proportion of wasted children to the decrease in *CICN* was different. In Maharashtra, almost 40 per cent of the decrease in *CICN* is attributed to the increase in the proportion of wasted children while this proportion is around 27 per cent in Dadra & Nagar Haveli and Daman & Diu but only around 13 per cent in Karnataka. On the other hand, almost one third of the decrease in *CICN* in Karnataka is attributed to the increase in the proportion of stunted children but this proportion is only about 20 per cent in Dadra & Nagar Haveli and Daman & Diu. In Maharashtra, on the other hand, increase in the proportion of stunted children accounted for only around 2 per cent of the decrease in *CICN*.

During the period 2019-2024, the progress in child nutrition reversed in 9 states/Union Territories as *CICN* decreased in these states/Union Territories. There is, however, no state/Union Territory where progress reversed in all the three dimensions of child nutrition. This is quite contrast to the period 2015-2021 when progress towards child nutrition was reversed in 14 states/Union Territories and there were 8 states/Union Territories progress reversed in all the three dimensions of child nutrition. The progress in child nutrition during the period 2019-2024 has been the most rapid in Nagaland where *CICN* increased from 0.736 during 2019-2021 to 0.844 during 2023-2024 whereas *CICN* decreased the most rapidly in Chandigarh from 0.817 during 2019-2021 to 0.765 during 2023-2024 because of the increase in the proportion of underweight children and the proportion of wasted children as the proportion of the stunted children decreased in the Union Territory during this period. The decrease in *CICN* during this period has also been quite rapid in Haryana and Punjab because of the increase in the proportion of underweight children and the proportion of wasted children as the proportion of stunted children decreased in both states. Chandigarh, Haryana and Punjab constitute a geographical contiguous region which is known for its prosperity and wealth and an advanced standard of living. It is surprising that despite the prosperity and advanced standard of living the proportion of underweight children and the proportion of wasted children in the region increased, instead decreased in the recent past.

Urban-Rural Difference

The proportion of children low height-for-age, low weight-for-height and low weight-for-height has been found to be low in the urban areas of the country as compared to its rural areas so that *CICN* is higher in the urban areas which implies that the nutritional status of children is better in the urban areas as compared to rural areas (Table 5). This is, however, not the case in four states – Nagaland, Puducherry, Punjab and Sikkim – where nutritional status of children appears to be better in the rural areas than in urban areas as *CICN* is higher in the rural areas compared to that in urban areas in these states and Union Territories. In states and Union Territories where *CICN* is higher in the urban areas, the urban-rural difference in *CICN* is found to be the widest in Dadra & Nagar Haveli and Daman & Diu followed by Jharkhand, Gujarat, Karnataka and Mizoram. But the narrowest in Tamil Nadu followed by Uttarakhand. Tamil Nadu is the only state/Union Territory in the country where there appears virtually little difference in the nutritional status of urban and rural children as reflected through the marginal difference in *CICN* between urban children and rural children of the state. In Sikkim, on the other hand, the nutritional status of rural children is markedly better than that of urban children as *CICN* is estimated to be substantially higher in the rural children as compared to the urban children of the state.

Table 6 decomposes the urban-rural difference in *CICN* into the urban-rural difference in the proportion of children who are not stunted, not underweight and not wasted. In most of the states/Union Territories, the urban-rural difference in all the three dimensions of child nutrition is the same as the urban-rural difference in *CICN*, although the relative contribution of the urban-rural difference in any dimension to the urban-rural difference in *CICN* varies across states/Union Territories. In Andhra Pradesh and Uttarakhand, around two third of the urban-rural difference in *CICN* is attributed to the urban-rural difference in the proportion of children who are not stunted. In Andhra Pradesh, there is no urban-rural difference in the proportion of children who are not wasted while there is no urban-rural difference in the proportion of children who are not underweight. In Bihar, Kerala and Dadra & Nagar Haveli and Daman & Diu, more than half of the urban-rural difference in *CICN* is attributed to the urban-rural difference in the proportion of children who are underweight. In Goa, on the other hand, more than 60 per cent of the urban-rural difference in *CICN* is due to the urban-rural difference in the proportion of children who are not low weight-for-height (wasted).

There are nine states/Union Territories in which the direction of the urban-rural difference in one of the three dimensions of child nutrition is not the same as the direction in the urban-rural difference in *CICN*. In seven states/Union Territories, the direction of the urban-rural difference in the proportion of children who are not wasted is opposite to the direction of urban-rural difference in *CICN*. In Lakshadweep, the direction of the urban-rural difference in the proportion of children who are not underweight is opposite to the direction of the urban-rural difference in *CICN* whereas the direction of the urban-rural difference in the proportion of children not stunted is opposite to the direction of the urban-rural difference in *CICN*. In four of these nine states/Union Territories, rural *CICN* is higher than the urban *CICN* which indicates that the nutritional status of rural children is relatively better than the nutritional status of urban children.

Table 5: Proportion of children who are low height-for-age, low weight-for-age, low weight-for-height and *CICN* in urban and rural areas of India and States/Union Territories, 2023-2024.

Country/State/Union Territory	Urban				CICN	Rural			CICN
	Proportion of children who are low			Proportion of children who are low					
	Height-for-age	Weight-for-age	Weight-for-height	Height-for-age		Weight-for-age	Weight-for-height		
India	23.9	25.3	16.5	0.780	30.9	33.8	19.8	0.716	
Andaman & Nicobar Islands	20.4	15.3	13.6	0.835	32.0	22.8	9.0	0.783	
Andhra Pradesh	21.7	21.8	12.3	0.813	25.8	23.8	12.3	0.792	
Arunachal Pradesh	23.5	16.0	14.2	0.820	24.4	19.3	17.1	0.797	
Assam	21.7	20.1	17.4	0.803	31.0	31.6	21.8	0.718	
Bihar	32.7	27.3	16.2	0.744	35.8	36.5	19.3	0.691	
Chandigarh	18.7	31.4	19.4	0.767					
Chhattisgarh	21.7	29.3	19.3	0.765	28.5	35.6	21.4	0.713	
Dadra & Nagar Haveli Daman & Diu	28.7	29.3	19.2	0.742	45.0	52.8	28.9	0.571	
Delhi	19.3	11.6	8.2	0.869	21.9	15.1	11.1	0.839	
Goa	18.8	19.5	14.7	0.823	20.0	20.5	18.3	0.804	
Gujarat	28.2	27.2	15.5	0.762	39.5	40.5	23.0	0.653	
Haryana	23.8	25.0	14.4	0.788	26.9	30.8	17.6	0.747	
Himachal Pradesh	16.4	11.8	4.4	0.890	20.9	17.3	11.0	0.835	
Jammu & Kashmir	19.3	11.6	8.2	0.869	21.9	15.1	11.1	0.839	
Jharkhand	20.3	29.4	20.4	0.765	37.4	43.1	22.6	0.652	
Karnataka	19.0	20.2	16.3	0.815	32.0	33.4	20.5	0.712	
Kerala	18.8	15.6	10.5	0.850	21.3	19.8	11.3	0.824	
Ladakh	22.1	12.3	8.2	0.856	26.7	15.0	11.1	0.822	
Lakshadweep	31.0	26.6	10.6	0.769	37.6	25.1	14.4	0.738	
Madhya Pradesh	24.5	30.5	19.3	0.751	33.2	42.0	24.9	0.664	
Maharashtra	24.9	26.5	18.3	0.767	32.2	34.3	20.9	0.707	

Country/State/Union Territory	Urban			CICN	Rural			CICN
	Proportion of children who are low				Proportion of children who are low			
	Height-for-age	Weight-for-age	Weight-for-height		Height-for-age	Weight-for-age	Weight-for-height	
Meghalaya	26.0	23.6	15.6	0.782	38.2	25.5	12.9	0.739
Mizoram	18.3	8.1	6.5	0.889	32.6	19.8	9.5	0.789
Nagaland	25.4	13.5	8.4	0.840	25.0	11.8	9.0	0.845
Odisha	18.5	22.1	22.6	0.789	28.0	32.9	22.1	0.722
Puducherry	26.4	27.4	15.1	0.769	28.5	26.6	10.3	0.779
Punjab	23.0	23.7	17.2	0.787	18.9	23.6	18.4	0.797
Rajasthan	21.8	22.5	14.8	0.802	31.4	35.8	20.9	0.704
Sikkim	18.2	21.0	8.0	0.841	17.8	8.6	9.8	0.879
Tamil Nadu	20.2	22.5	17.9	0.798	21.0	23.6	17.1	0.794
Telangana	22.3	24.3	15.2	0.793	28.6	28.9	17.4	0.749
Tripura	18.5	23.7	16.9	0.803	22.4	26.0	18.7	0.776
Uttar Pradesh	28.8	32.0	18.7	0.733	32.1	35.0	19.4	0.709
Uttarakhand	17.7	19.6	9.7	0.843	20.7	19.6	11.4	0.827
West Bengal	19.3	21.9	14.2	0.815	23.3	30.3	22.0	0.747

Remarks: NFHS-6 was not conducted in Manipur.

Source: Author

Table 6: Decomposition of the urban-rural difference in *CICN*, 2023-2024.

Country/State/Union Territory	Urban-rural difference in <i>CICN</i>	Urban-rural difference in <i>CICN</i> attributed to the urban-rural difference in the proportion of children not low		
		Height-for-age	Weight-for-age	Weight-for-height
India	0.064	0.024	0.030	0.010
Andaman & Nicobar Islands	0.052	0.041	0.025	-0.014
Andhra Pradesh	0.021	0.014	0.007	0.000
Arunachal Pradesh	0.023	0.003	0.011	0.009
Assam	0.085	0.032	0.039	0.014
Bihar	0.052	0.011	0.032	0.009
Chhattisgarh	0.052	0.022	0.022	0.007
Dadra & Nagar Haveli Daman & Diu	0.170	0.056	0.085	0.029
Delhi	0.030	0.009	0.012	0.009
Goa	0.019	0.004	0.003	0.012
Gujarat	0.109	0.040	0.047	0.023
Haryana	0.041	0.011	0.020	0.010
Himachal Pradesh	0.055	0.016	0.018	0.021
Jammu & Kashmir	0.030	0.009	0.012	0.009
Jharkhand	0.113	0.057	0.049	0.007
Karnataka	0.103	0.044	0.045	0.013
Kerala	0.025	0.009	0.014	0.003
Ladakh	0.035	0.017	0.009	0.009
Lakshadweep	0.031	0.024	-0.005	0.011
Madhya Pradesh	0.088	0.029	0.041	0.017
Maharashtra	0.060	0.025	0.027	0.008
Meghalaya	0.042	0.044	0.006	-0.008
Mizoram	0.100	0.052	0.038	0.009
Nagaland	-0.005	-0.001	-0.006	0.002
Odisha	0.067	0.031	0.037	-0.002
Puducherry	-0.010	0.007	-0.003	-0.015
Punjab	-0.010	-0.014	0.000	0.004
Rajasthan	0.098	0.033	0.046	0.019
Sikkim	-0.037	-0.001	-0.042	0.006
Tamil Nadu	0.004	0.003	0.004	-0.003
Telangana	0.044	0.022	0.016	0.007
Tripura	0.027	0.013	0.008	0.006
Uttar Pradesh	0.024	0.011	0.011	0.002
Uttarakhand	0.016	0.010	0.000	0.005
West Bengal	0.068	0.013	0.029	0.025

Remarks: There is no rural population in Chandigarh.

NFHS-6 was not conducted in Manipur.

Source: Author

Discussion and Conclusion

This paper has followed the concept of child nutrition surface and the associated composite metric of child nutrition (*CICN*) first proposed by Chaurasia (2025) to measure and monitor the progress in child nutrition in India and in its 35 states and Union Territories during 2015-2024. The concept of child nutrition surface recognises that nutritional status of children is a multi-dimensional construct with linear growth, ponderal growth and body mass growth its three core dimensions. A metric based on the child nutrition surface, therefore, is more appropriate to rank and compare populations and to monitor the progress in child nutrition than metrics which capture nutritional status of children in the population in individual dimensions of child nutrition only. *CICN* presents a broad, macro perspective of child nutrition in the population that reflects progress in the three dimensions of child nutrition simultaneously.

Assessing the nutritional status of children in terms of *CICN* using the evidence available from the latest round of the National Family Health Survey (NFHS-6) suggests that the problem of child undernutrition in India continues to be a challenge in the development discourse of the country. Although, the survey reveals progress in child nutrition in the country, yet the progress remains slow. If the improvement in *CICN* observed during the period 2019-2024 is any indication, then it will take at least three decades for the country to achieve a satisfactory level of the nutritional status of its children. Even more intriguing is the findings of the present analysis that progress in child nutrition has been inconsistent within the country, across states/Union Territories. Child nutrition has not improved in all states and Union Territories during the period under reference. The evidence from the latest NFHS indicates that the progress in child nutrition appears to have reversed in the recent past in many states and Union Territories. Another intriguing revelation of the present analysis is that states/Union Territories in which progress in child nutrition reversed during the period 2015-2021 are different from states/Union Territories in which progress reversed during the period 2019-2024. There is only one state/Union Territory – Dadra & Nagar Haveli and Daman & Diu – where progress in child nutrition reversed throughout the period 2015-2024. The inconsistency in the progress in child nutrition in many states and Union Territories of the country raises questions about the appropriateness, efficiency and effectiveness of the efforts directed towards improving the nutritional status of children.

The evidence available from different rounds of NFHS also suggests that the progress in the three dimensions of child nutrition in the country is highly imbalanced. In India and in most of its states and Union Territories, progress in child nutrition has been limited to the linear growth dimension of child nutrition whereas there has been little progress in other dimensions of child nutrition. The analysis suggests that more than 90 per cent of the progress in child nutrition in the country during 2015-2024 as captured through the increase in *CICN* is attributed to the increase in the proportion of children who are not low height-for-age or stunted, although there are many states and Union Territories in which progress reversed in this dimension of child nutrition also. This highly imbalanced progress in different dimensions of child nutrition appears to be a factor behind the slow progress in improving the nutritional status of children. It also indicates that efforts to address the challenge of child undernutrition in the country has, at best, been lop-sided. Linear growth faltering is an indicator of chronic undernutrition while ponderal growth faltering is an indicator of acute undernutrition and the strategies and approaches required

to address linear growth faltering and ponderal growth faltering are essentially different. It appears that the focus of efforts directed towards improving the nutritional status of children is on preventing linear growth faltering while efforts necessary to check ponderal growth faltering have received a neglected attention. The marginal progress in the ponderal growth dimension has also contributed to the marginal progress in the body mass growth dimension of child nutrition in the country.

Efforts to address the challenge of child undernutrition in India are currently being organised under the National Nutrition Mission, also known as the Poshan Abhiyaan or Nutrition Movement. This Mission was launched by the Government of India as a centrally sponsored programme in 2018 (Government of India, 2018). The Mission has the ambitious goal of improving nutritional outcomes among adolescent girls, pregnant and lactating women, and children under six years of age through technology, convergence, and community involvement. The evidence available from the latest round of the National Family Health Survey (NFHS-6), however, indicates that the Mission appears to have overlooked the local context of child nutrition so that the progress in child nutrition varies widely across states and Union Territories of the country. Since the local context of child nutrition varies widely across states and Union Territories and even across districts within states and Union Territories, it is imperative that a decentralised approach should have been adopted by the Mission to plan, implement and monitor the progress efforts directed towards improving the nutritional status of children. There is a need to examine the needs effectiveness and capacity efficiency of the National Nutrition Mission in addressing the local context of child nutrition. Such an exercise may preferably be carried out at the district level where all child nutrition related activities are actually implemented. District level estimates of the proportion of children below five years of age who are low height-for-age (stunted), low weight-for-age (underweight) and low height-for-weight (wasted) based on NFHS, 2023-2024 have recently been released (Government of India, 2026). These estimates may be used for ranking districts and for exploring the factors responsible for inter-district variation in child undernutrition in the country.

References

- Ang BW, Zhang FQ, Choi Ki-Hong (1998) Factorizing changes in energy and environmental indicators through decomposition. *Energy* 23(6): 489-495.
- Ang BW, Liy Na (2007) Handling zero values in the logarithmic mean Divisia index decomposition approach. *Energy Policy* 35(1): 238-246.
- Ang BW (2015) LMDI decomposition approach: a guide for implementation. *Energy Policy*, 86: 233-238.
- Al-Sadeeq AH, Bukair AZ, Al-Saqladi AM (2018) Assessment of undernutrition using Composite Index of Anthropometric Failure among children aged <5 years in rural Yemen. *Eastern Mediterranean Health Journal* 24(12)
- Benjamin-Chung J, Mertens A, Colford Jr JM, Hubbard AE, van der Laan MJ, Coyle J, Sofrygin O, Cai, Nguyen A, Pokpongkiat NN, Djajadi S, Seth A, Jilek W, Jung E, Chung EO, Rosete S, Hejazi N, Malenica I, Li H, Hafen H, Subramoney V, Häggström J, Norman

- T, Brown KH, Christian P, Arnold BF and The Ki Child Growth Consortium (2023) Early-childhood linear growth faltering in low- and middle-income countries. *Nature* 621: 550-557.
- Bose K, Mandal GC (2010) Proposed new anthropometric indices of childhood undernutrition. *Malaysian Journal of Nutrition* 16(1): 131-136.
- Carlson BC (1974) The logarithmic mean. *The American Mathematical Monthly* 79(6): 615-618.
- Chaurasia AR (2017) Decomposition of body mass growth into linear and ponderal growth in children with application to India. *British Journal of Nutrition* 117(3): 413-421.
- Chaurasia AR (2025) A composite index of child nutrition with application to India. *Social Indicators Research*. <https://doi.org/10.1007/s11205-025-03676-w>.
- Gausman G, Kim R, Li Z, Tu L, Rajpal S, Joe W, Subramanian SV (2022) Comparison of child undernutrition anthropometric indicators across 56 low- and middle-income countries. *JAMA Network Open* 5(3): e221223. Government of India
- Government of India (2017) *National Family Health Survey (NFHS-4), 2015–16: India*. Mumbai, International Institute for Population Sciences and ICF.
- Government of India (2020) *Accelerating Progress on Nutrition in India. What Will It Take? Third Progress Report*. New Delhi, NITI Aayog.
- Government of India (2022) *National Family Health Survey (NFHS-5), 2019–21: India*. Mumbai, International Institute for Population Sciences and ICF.
- Government of India (2026a) *National Family Health Survey (NFHS-6), 2023–24. Fact Sheets*. Mumbai, International Institute for Population Sciences.
- Government of India (2026b) *National Family Health Survey (NFHS-6), 2023–24. Compendium of Fact Sheets. Key Indicators of Districts*. Mumbai, International Institute for Population Sciences.
- Isanaka S, Hitchings MDT, Berthé F, Briend A, Graiss RF (2019) Linear growth faltering and the role of weight attainment: prospective analysis of young children recovering from severe wasting in Niger. *Maternal and Child Nutrition* 15(4):e12817.
- Mertens A, Benjamin-Chung J, Colford Jr JM, Coyle J, van der Laan MJ, Hubbard AE, Rosete S, Malenica I, Hejazi N, Sofrygin O, Cai W, Li H, Nguyen A, Pokpongkiat NN, Djajadi S, Seth A, Jung E, Chung EO, Jilek W, Subramoney V, Hafen R, Häggström J, Norman T, Brown KH, Christian P, Arnold BF, The Ki Child Growth Consortium (2023) Causes and consequences of child growth faltering in low-resource settings. *Nature* 621:568-576.
- Nandy S, Irving M, Gordon D, Subramanian SV, Smith GD (2005) Poverty, child under nutrition and morbidity: new evidence from India. *Bulletin of World Health Organization* 83: 210–216.
- Svedberg P (2000) *Poverty and Under-nutrition: Theory, Measurement and Policy*. Oxford, Oxford University Press.

CHAURASIA; IJPD 6(1): 1-22

Taylor BN, Kuyatt CE (1994) Guidelines for evaluating and expressing the uncertainty of NIST measurement results. Gaithersburg, National Institute of Standards and Technology. Physics Laboratory. NIST Technical Note 1297.

United Nations (2024) *World Population Prospects 2024. Online Edition*. New York, Department of Economic and Social Affairs, Population Division.

Wood R, Lenzen M (2006) Zero-value problems of the logarithmic mean division index decomposition method. *Energy Policy* 34(12): 1326-1331.